ModelIt!

Teacher's Guide

Building a Star on Earth: The Fusion Energy Challenge

Modeling the Engineering Extremes of Controlled Nuclear Fusion

Quick Reference

Grade Level	Grade 10L3
NGSS Standard	HS-ESS1-1, HS-PS3-5
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-ESS1-1, HS-PS3-5

Develop a model based on evidence to illustrate the life span of the sun and the role of nuclear fusion in the sun's core to release energy. Plan and conduct an investigation to provide evidence that an electric current can produce a magnetic field and that a changing magnetic field can produce an electric current.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop computational models to illustrate how nuclear fusion processes in stellar cores translate to engineering challenges in terrestrial reactor design.
Disciplinary Core Idea	**ESS1.A The Universe and Its Stars / PS3.D Energy in Chemical Processes and Everyday Life** — Students model how nuclear fusion releases energy through mass-energy conversion and investigate the engineering systems needed to capture and convert that energy.
Crosscutting Concept	**Energy and Matter** — Students trace energy flows from nuclear binding energy through plasma heating, neutron capture, thermal transfer, and electrical conversion, quantifying losses at each stage.

CAST Testing Alignment

MULTIPLE CHOICE:

A fusion reactor achieves a plasma temperature of 150 million degrees Celsius but the magnetic containment field fluctuates by 15%. According to your model, what is the most likely immediate consequence?

CONSTRUCTED RESPONSE:

Using your model, explain why achieving Q greater than 1 (net energy gain) requires solving plasma physics, materials science, and engineering challenges simultaneously. Why can't scientists solve them one at a time?

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

The Sun converts 600 million tons of hydrogen into helium every second, releasing energy equivalent to 10 billion nuclear bombs. This process — nuclear fusion — has powered every star in the universe for billions of years. Recreating it on Earth would provide virtually unlimited clean energy. But building a star is, unsurprisingly, extraordinarily difficult.

How Fusion Works

Nuclear fusion occurs when light atomic nuclei combine to form heavier ones, releasing energy from the strong nuclear force. The most achievable reaction combines deuterium (hydrogen with one neutron) and tritium (hydrogen with two neutrons) to produce helium-4, a high-energy neutron, and 17.6 MeV of energy. But nuclei are positively charged and repel each other — overcoming this Coulomb barrier requires temperatures exceeding 150 million degrees Celsius, ten times hotter than the Sun's core. At these temperatures, matter exists as plasma.

The Confinement Problem

No physical container can hold 150-million-degree plasma — it would vaporize instantly. Instead, fusion reactors use magnetic confinement: powerful superconducting magnets create a magnetic bottle that suspends the plasma in a vacuum. The most common design is the tokamak, a donut-shaped chamber where magnetic field lines spiral around the plasma. But plasma is inherently unstable — turbulence, instabilities, and edge effects constantly try to break confinement, requiring real-time feedback control systems.

The Materials Crisis

Every deuterium-tritium fusion reaction produces a 14.1 MeV neutron that escapes the magnetic field (neutrons are uncharged). These neutrons slam into reactor wall materials, displacing atoms from their crystal lattice in a process called radiation damage. Over years of operation, structural materials become brittle, swollen, and radioactive. No existing material can withstand the neutron flux of a commercial fusion reactor for its intended 30-year lifetime. This materials challenge is arguably the biggest unsolved problem in fusion engineering.

The Race to Net Energy

In December 2022, the National Ignition Facility achieved fusion ignition for the first time — producing more energy from fusion than the laser energy used to trigger it (Q greater than 1). But this was using inertial confinement (lasers), not magnetic confinement, and the total facility energy consumption was far higher than the fusion output. ITER, the international tokamak being built in France, aims to achieve Q=10 by the early 2030s. Private companies like Commonwealth Fusion Systems and Helion Energy are pursuing faster timelines with innovative designs. The race to commercial fusion is accelerating.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify nine critical components of a fusion reactor system: plasma temperature, magnetic containment, fuel injection rate, energy output, neutron damage, cooling system, power conversion, materials degradation, and net energy gain.

E **Establish Relationships**

Students map relationships showing that increasing plasma temperature and fuel injection boosts energy output but simultaneously accelerates neutron damage and materials degradation — revealing the fundamental tension in fusion engineering.

V **Visualize & Model**

Students build a computational model showing the nine-component fusion system with trade-offs between energy production and reactor longevity.

E **Evaluate Outcomes**

Students run ITER baseline, ignition attempt, and containment instability scenarios to identify the operating envelope where net energy gain is achievable without destroying the reactor.

R **Revise & Extend**

Students add tritium breeding ratio, plasma disruption frequency, or superconductor temperature to model more complete fusion reactor dynamics.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"The Sun does it effortlessly — 600 million tons of hydrogen fused every second for 4.6 billion years. We have spent 70 years and 50 billion dollars trying to do the same thing in a building. Why is it so hard?"*

Watch For

Show a side-by-side of the Sun's core and a tokamak interior. Ask: What does the Sun have that we do not?

Slide 2: Learning Objectives

3 min

Say: *"Today you are modeling the most extreme engineering challenge on Earth — temperatures ten times hotter than the Sun, magnets cooled to near absolute zero, and materials that must survive neutron bombardment for decades."*

Watch For

Have students read objectives. Pre-teach 'plasma' and 'magnetic confinement.' Use the temperature contrast: 150 million degrees plasma, 3 meters from -250 degree magnets.

Slide 3: Big Question

3 min

Say: *"Fusion promises unlimited clean energy with no greenhouse gases and no long-lived radioactive waste. So why does it not exist yet? What is the engineering barrier?"*

Watch For

Quick-write: Students list what they think are the biggest challenges. Share out. Track on board — revisit at the end to see which ones the model confirms.

Slide 4: LEVER Framework

2 min

Say: *"We are modeling a system where every variable is at a physical extreme. Temperature, pressure, magnetic fields, neutron flux — all pushed to limits that do not exist anywhere else on Earth."*

Watch For

Preview LEVER steps. Emphasize the nine-component model and the fundamental trade-off between output and damage.

Slide 5: Activity 1: Components

8 min

Say: *"Nine variables, three inputs you control, six outputs that respond. The challenge: the outputs you WANT (energy) come packaged with outputs you do NOT want (damage)."*

Watch For

Guide students through components. Discuss why temperature, containment, and fuel injection are external — they are the engineering inputs. Everything else is a consequence.

Slide 6: Activity 2: Connections

8 min

Say: *"Here is the cruel paradox of fusion: every reaction that produces energy also produces a neutron that is destroying your reactor. More power means more damage. How do you optimize that?"*

Watch For

Help students map the trade-off relationships. Highlight the output-damage coupling. Use red arrows for destructive pathways and green for productive ones.

Slide 7: Activity 3: Simulation

10 min

Say: *"Can your reactor achieve Q greater than 10? Let us push the parameters and find out — but watch what happens to your reactor walls."*

Watch For

Students predict Net Energy Gain before running each scenario. Compare predictions with simulation results. Discuss the sweet spot between output and longevity.

Slide 8: Discoveries

5 min

Say: *"Now you understand why fusion has been 30 years away for the last 60 years. It is not that we cannot make fusion happen — it is that we cannot make it happen economically while keeping the reactor alive."*

Watch For

Lead discussion connecting model to real-world fusion milestones. NIF ignition, ITER timeline, private fusion companies. Where is the breakthrough coming from?

Slide 9: STEM Challenge

12 min

Say: *"The DOE wants your ideas. What is the biggest bottleneck, and how would you break through it?"*

Watch For

Groups identify their chosen bottleneck and design innovative solutions. Present with evidence from model simulations. Class evaluates feasibility.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Plasma Temperature, Magnetic Containment, Fuel Injection Rate
- **INTERNAL:** Energy Output, Neutron Damage, Cooling System, Power Conversion, Materials Degradation, Net Energy Gain

Relationships

- **Plasma Temperature to Energy Output:** POSITIVE (+) — Higher plasma temperature increases the rate of fusion reactions exponentially (following the fusion cross-section curve), directly increasing energy output. The reaction rate scales strongly with temperature above the ignition threshold.
- **Energy Output to Neutron Damage:** POSITIVE (+) — Higher energy output means more fusion reactions per second, and each reaction produces a 14.1 MeV neutron. More fusion events mean more neutrons bombarding reactor walls, accelerating structural damage to the first wall and blanket materials.
- **Neutron Damage to Materials Degradation:** POSITIVE (+) — Accumulated neutron damage displaces atoms from their crystal lattice positions, creating voids, helium bubbles, and transmutation products. Over time, this makes structural materials brittle, swollen, and prone to cracking, ultimately limiting reactor operational lifetime.

Key Discoveries

- Fusion requires simultaneously achieving extreme temperature, density, and confinement time — the Lawson criterion — and falling short on any single parameter means zero net energy
- Every fusion reaction that produces energy also produces a high-energy neutron that damages the reactor, creating a fundamental tension between output and longevity
- Magnetic containment is inherently unstable — plasma turbulence constantly tries to escape confinement, requiring real-time feedback control systems adjusting thousands of times per second
- Net energy gain requires not just producing fusion energy but producing ENOUGH to overcome the enormous energy cost of heating plasma, powering magnets, and running cooling systems

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Fusion reactors can explode like nuclear bombs"	A fusion reactor cannot explode like a thermonuclear weapon. Weapons require precisely shaped implosion to compress fuel to extreme density in nanoseconds — conditions impossible to achieve accidentally. If a fusion reactor loses containment, the plasma cools in milliseconds and fusion stops. The total fuel in a reactor at any moment is about one gram — there is no chain reaction to sustain. A fusion failure is a very expensive equipment malfunction, not a nuclear explosion.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"We just need hotter plasma to make fusion work"	Temperature is necessary but not sufficient. The Lawson criterion requires high temperature AND high density AND long confinement time — simultaneously. Many experiments achieve extreme temperatures but cannot maintain confinement long enough or at sufficient density. It is like saying 'we just need more flour to make bread' — you also need water, yeast, heat, and time, all in the right proportions.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Fusion produces no radioactive waste"	Fusion produces far less radioactive waste than fission and no long-lived actinides, but it is not waste-free. The 14.1 MeV neutrons from fusion reactions activate structural materials through neutron capture, creating radioactive isotopes. These materials must be stored for 50-100 years before they decay to safe levels — much shorter than fission waste's thousands of years, but not zero. Tritium is also radioactive with a 12.3-year half-life and must be carefully contained.